FIITJEE

JEE (Advanced), 2018

ALL INDIA TEST SERIES

FULL TEST – I

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 234**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section-A (01 – 08)** contains 8 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **–1 mark** for wrong answer.

   Section-A (09 – 12) contains 4 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

   Section-A (13 to 18) contains 3 paragraphs with each having 2 questions. Each question carries **+3 marks** for correct answer and **–1 mark** for wrong answer.

2. **Section-C (01 – 05)** contains 5 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

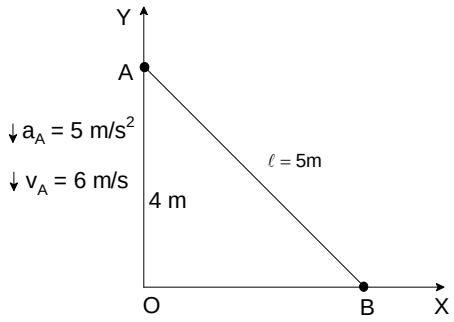
Physics

PART – I

SECTION – A
Straight Objective Type

This section contains **08 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct.

1. Point A on the rod AB has an acceleration of 5 m/s^2 and a velocity of 6 m/s at an instant as shown in the figure. The acceleration of the end B at the same moment is:



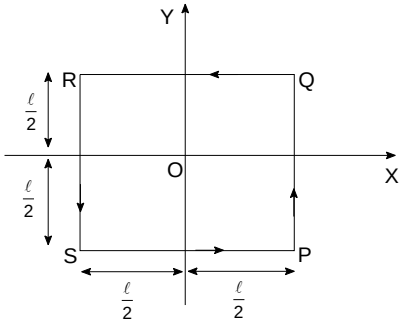
(A) $-\frac{80}{3}\hat{i}$ m/s^2

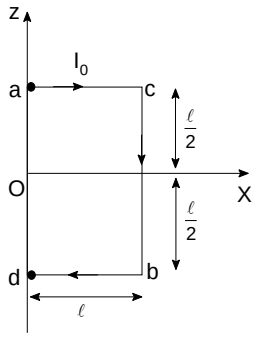
(B) $\frac{80}{3}\hat{i}$ m/s^2

(C) $-\frac{40}{3}\hat{i}$ m/s^2

(D) $-\frac{75}{6}\hat{i}$ m/s^2

2. Current I_0 is flowing through a bent wire $a \to b \to c \to d$ in z – x plane as shown in figure, then $\oint \vec{B}.\vec{d\ell}$ over the loop PQRS lying in the X-Y plane as shown in figure, due to the bent wire abcd





(A) $\frac{\mu_0 I_0}{8}$

(B) $\frac{\mu_0 I_0}{7}$

(C) $\frac{\mu_0 I_0}{5}$

(D) $\frac{\mu_0 I_0}{3}$

Space for rough work

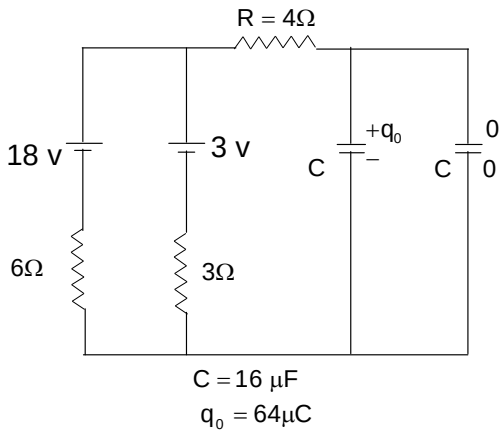
3. The circuit is completed at t = 0, then heat loss in $R = 4\Omega$ resistance, till the capacitor gets fully charged:

(A) $64\,\mu J$

(B) $384\,\mu J$

(C) $320\,\mu J$

(D) $576\,\mu J$



4. A simple harmonic plane progressive wave is travelling along the +ve x – axis having amplitude A and $\omega = \pi$ rad / s. At t = 1 sec, particle at x = 2 m and x = 3 m are moving in downward and upward direction respectively at the position $y = +\frac{A}{2}$. The time when maximum power transfer will takes place at $x = \frac{11}{4}$m?

(A) $\frac{5}{4}$sec

(B) $\frac{5}{3}$ sec

(C) $\frac{7}{4}$ sec

(D) $\frac{7}{5}$ sec

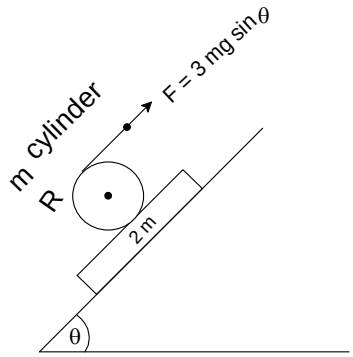
5. There is sufficient friction between long plank of mass 2 m and cylinder of mass m and radius R to prevent slipping. Whole system is placed on the smooth inclined surface. Now, force $F = 3mg\sin\theta$ is applied on the ideal string wrapped over the cylinder as shown in figure. Choose correct option:

(A) If the plank moves down a distance of S m, work done by the applied force F on the string is 5 mg $\sin\theta$ S

(B) if the plank moves down a distance of S m, work done by the applied force F on the string is 6 mg $\sin\theta$ S

(C) If the plank moves down a distance of S m, work done by the applied force F on the string is 12 mg $\sin\theta$ S

(D) If the plank moves down a distance of S m, work done by the applied force F on the string is 15 mg $\sin\theta$ S



Space for rough work

6. A vertical thermally insulated cylinder of volume V contain n moles of an ideal monoatomic gas under a weightless piston. A load of weight W is placed on the piston as a result of which the piston is displaced. If the initial temp of the gas is 300 K, area of piston is A and atmospheric pressure P_0. (take $W = P_0A$). Determine the value of final temperature of the gas.
(A) 375 K (B) 425 K
(C) 475 K (D) 515 K

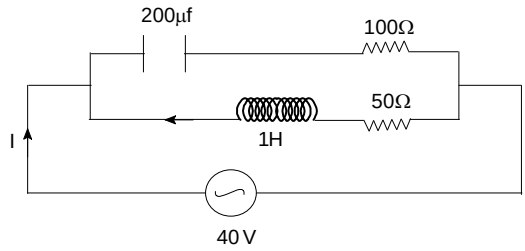
7. A open organ pipe containing air resonates in fundamental mode due to a tuning fork. The measured value of length $\ell(\text{in cm})$ of the pipe and radius $r(\text{in cm})$ of the pipe are $\ell = 94.0 \pm 0.1, r = 5.0 \pm 0.05$. The velocity of the sound in air is accurately known. The maximum percentage error in the measurement of the frequency of that tuning fork by thus experiment, will be:
(A) 0.16 (B) 0.64
(C) 1.2 (D) 1.6

8. A metallic spherical shell has mass m, radius R and Bulk modulus of elasticity B. The change in radius of the shell due to gravitational pressure
(A) $\frac{Gm^2}{3B\pi R^3}$ (B) $\frac{Gm^2}{8B\pi R^3}$
(C) $\frac{Gm^2}{24B\pi R^3}$ (D) $\frac{Gm^2}{36B\pi R^3}$

Multiple Correct Choice Type

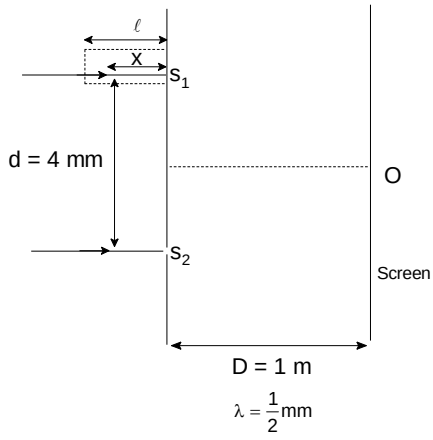
This section contains **04 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

9. In the given circuit, the AC source has $\omega = 50 \text{ rad/s}$ considering the inductor and capacitor to be ideal, the correct choice(s) is/are
(A) The voltage across 100Ω resistor $= 20$
(B) The voltage across 50Ω resistor $= 20$
(C) The current through the circuit is $I = \frac{\sqrt{5}}{\sqrt{2}}$ A
(D) The current through the circuit is $I = 1.2$A



Space for rough work

10. In YDSE, D = 1 m, d = 4 mm and $\lambda = \frac{1}{2}$mm In the figure shown, a parallel beam of light is incident on the plane of the slits of YDSE. Light incident on the slit S_1, passes through a medium of variable refractive index $\mu = (2x + 1)$ (where x is the distance from the plane of slits in mm) up to a distance $\ell = 1$ mm before falling on S_1. Rest of the space filled with air. Then choose the correct option(s):



(A) Position of central bright fringe is at a distance

$y = \frac{1}{\sqrt{3}}$m above the central line

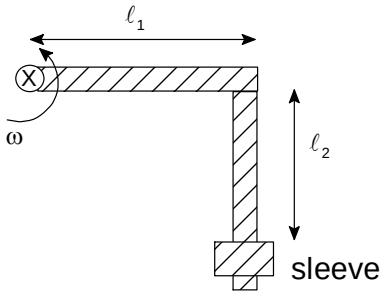
(B) Position of central bright fringe is at a distance

$y = \frac{1}{2}$m above the central line

(C) Number of maxima obtaining on the screening is 7

(D) Number of maxima obtaining on the screening is 8

11. A rough L – shaped rod is located in a horizontal plane and a sleeve of mass m is inserted in the rod. The rod is rotated with a constant angular velocity ω in the horizontal plane. The lengths ℓ_1 and ℓ_2 are shown in figure. The normal reaction and frictional force acting on the sleeve when it starts slipping are (μ : coefficient of friction between rod and sleeve).



(A) $N = m\omega^2\ell_1$

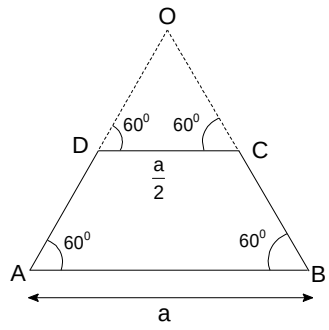
(B) $\omega^4 = \frac{\mu^2 g^2}{\ell_2^2 - \mu^2\ell_1^2}$

(C) $N = m\sqrt{g^2 + \omega^4\ell_1^2}$

(D) $f = \mu N$

Space for rough work

12. Consider a uniformly charged sheet ABCD, which is a part of an equilateral triangular sheet of a side a as shown in the figure. Choose the correct options regarding the electric field E at point O due to this sheet.



(A) Magnitude of $\vec{E}$, increases with the increase in a (Keeping charge density same)

(B) Magnitude of E, decreases with increase in a (keeping total charge same)

(C) If charge density is σ and a = 1 m, magnitude of E is equal to $\frac{7\sigma}{44\varepsilon_0}\ln\sqrt{2}$

(D) If charge density is σ and a = 2 m, magnitude of E is equal to $\frac{7\sigma}{22\varepsilon_0}\ln\sqrt{2}$

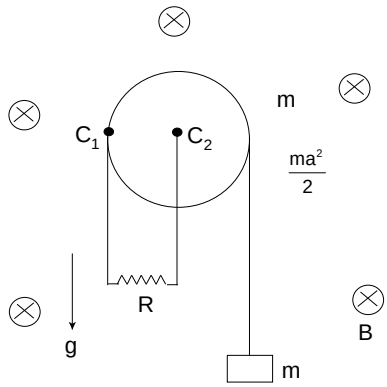
Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct

Paragraph for Questions 13 & 14

Consider a perfectly conducting uniform disc of mass m and radius 'a' hinged in a vertical plane from its centre and free to rotate with respect to hinge. A resistance R is connected between centre of the disc and periphery by using two sliding contacts C_1 & C_2. A long non – conducting massless string is wrapped around the disk, whose another end is attached with a block of mass m. There exists a uniform horizontal magnetic field B, whose arrangement is shown in figure.

Given system is released from rest at t = 0. Assume friction between string and disc is sufficient so that there is no slipping between them let at any instant t, velocity of block is v, angular velocity of the disc is ω and current in resistance is i.



13. Find the terminal speed of the block of mass m is (Terminal speed = constant speed attained by the block after very long time)

(A) $\frac{mgR}{B^2a^2}$ (B) $\frac{2mgR}{B^2a^2}$

(C) $\frac{3mgR}{B^2a^2}$ (D) $\frac{4mgR}{B^2a^2}$

Space for rough work

14. Just after t = 0, the acceleration of block is:

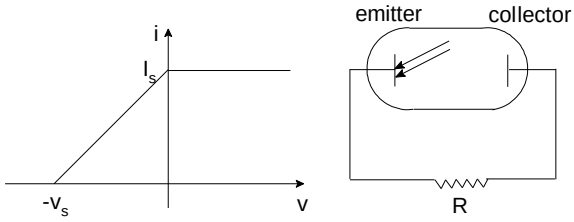
(A) g (B) $\frac{g}{2}$

(C) $\frac{2g}{3}$ (D) $\frac{g}{5}$

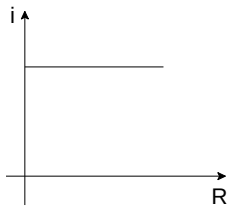
Paragraph for Questions 15 & 16

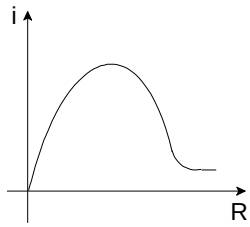
Let us assume that the current voltage graph for photoelectric tube consists of two straight lines as shown.

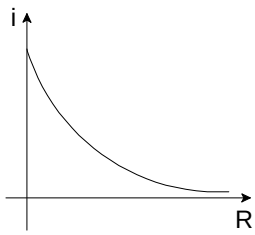
On such a photoelectric tube, light of wavelength 310 nm is incident. The work function of the emitter plate is 2eV. The tube effectively acts as a solar cell to supply current to an external resistor R. Saturation current is 2A.

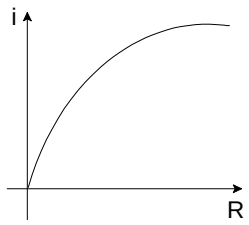


15. How does the current depend for R

(A)


(B)


(C)


(D)


Space for rough work

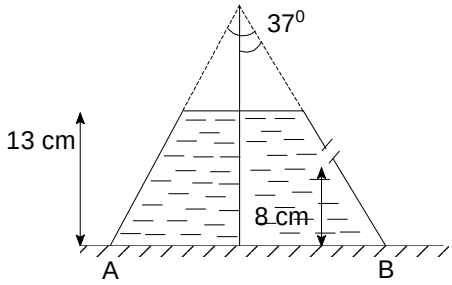
16. At the what value of R will be the power dissipated in R maximum

(A) 1Ω (B) 2Ω

(C) $\frac{1}{2}\Omega$ (D) $\frac{1}{4}\Omega$

Paragraph for Questions 17 & 18

Curved surface of a vessel has shape of truncated cone having semi – vertex angle 37^0. Vessel is full of water (density $\rho = 1000\ kg/m^3$) upto a height of 13 cm and is placed on a smooth horizontal plane. Upper surface is opened to atmosphere. A hole of 1.5 cm is made on curved wall of a height of 8 cm from bottom as shown in figure. Area of water surface in the vessel is large as compared to the area of the hole.



17. The initial horizontal force required to keep the vessel in static equilibrium

(A) 0.12N (B) 0.15 N

(C) 0.25 N (D) 0.35 N

18. The initial horizontal range of water from point B is

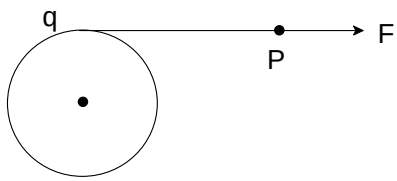
(A) 16 cm (B) 12 cm

(C) 10 cm (D) 8cm

SECTION – C
(One Integer Value Correct Type)

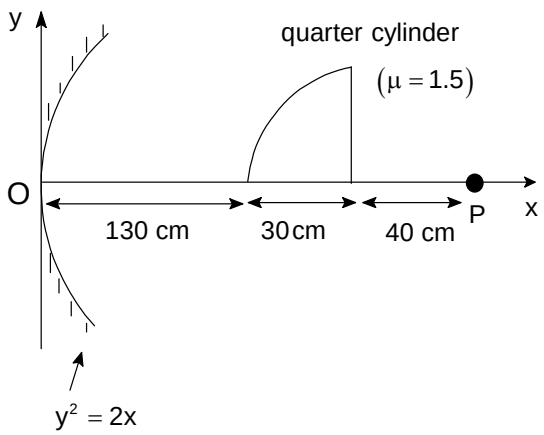
This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

19. A very long non- conducting cylindrical shell of length $\ell = 10m$ is free to rotate about its fixed axis. A light inextensible thread is wrapped over non – conducting cylindrical surface and one end P is pulled with constant acceleration of $\frac{F}{2m}$. F is the force applied at P and m is the mass of cylinder. Then charge on the cylinder will be $5\times10\ K$ coulomb. $[\text{Given } m = 25\ kg, \ell = 10m]$. Find the value of K?

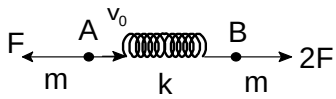


Space for rough work

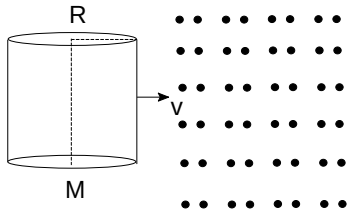
20. A point object is kept at a distance of 2 m from origin on x – axis. A quarter cylinder of radius R = 30 cm and refractive index $\mu = 1.5$ having its axis $\perp_r$ to plane of paper is placed between the object and the parabolic reflector at a distance of 130 cm from origin as shown in figure. The position of the image after reflection from the parabolic reflector $y^2 = 2x$ is 25 λ cm. Find the value of λ?



21. Two identical particles A & B of mass m = 2 kg is connected at the end of the spring $(K = 50\ N/m)$ and placed on a smooth horizontal surface. Now, force F = 10 N and 2 F are applied on the particles in opposite direction as shown in figure. Initially spring is in natural length.
Simultaneously particle A receives an impulse towards B which imparts speed v_0 to it, find v_0 in m/s for which length of elongation is two times the length of compression in the spring during subsequent motion?



22. A cylinder of mass M and radius R moves with constant speed v through a region of space that contains dust particles of mass m which is at rest. There are n number of particles per unit volume. The cylinder moves in a direction perpendicular to its axis. Assume m <<M, and assume the particles do not interact with each other. All the collision takes place is perfectly elastic and the surface of the cylinder is smooth. The drag force per unit length of the cylinder require to maintain the speed v constant for the cylinder is $\frac{K}{3}nmRv^2$. Find the value of K?



23. In a hydrogen like atom an electron is orbiting in an orbit having quantum number n. Its frequency of revolution is found to be 13.2×10^{15}Hz. Energy required to free the electron from the atom from the above orbit is 54.4 eV. In time 7 nano – second the electrons jumps back to orbit having quantum number $\frac{n}{2}$. τ be the average torque acted on the electron during the above process, then find $\frac{\tau}{5}\times10^{27}$ in N – m.

$\left(\text{Given } \frac{h}{\pi} = 2.1\times10^{-34}\ J-s\right)$ Frequency of revolution of electron in the ground state of H – atom $f_0 = 6.6\times10^{15}$Hz and ionization energy of H – bond $E_0 = 13.6$ eV

Space for rough work

Chemistry

PART – II

SECTION – A
Straight Objective Type

This section contains **08 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct.

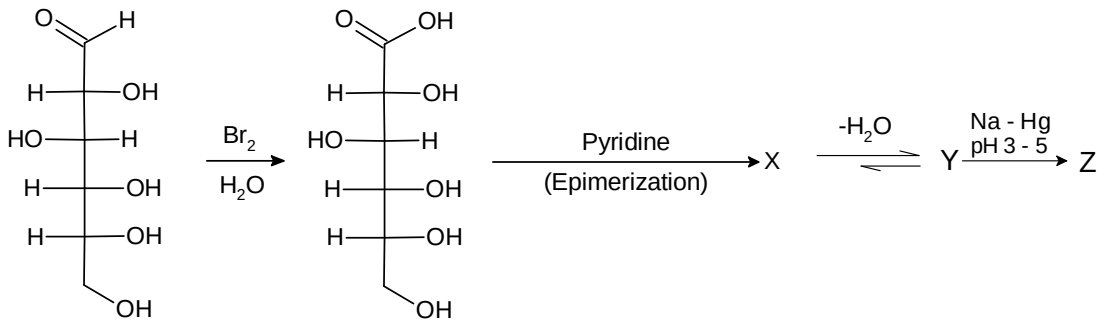
24. A gas X is passed through water to form a saturated solution. The aqueous solution on treatment with $AgNO_3$ gives white precipitate. The saturated aqueous solution also dissolves magnesium ribbon with evolution of a colourless gas Y.
(A) X is CO_2 and Y is Cl_2 (B) X is Cl_2 and Y is CO
(C) X is Cl_2 and Y is H_2 (D) X is H_2 and Y is Cl_2

25. The method of zone refining of metals is based on the principle of:
(A) Greater mobility of the pure metal than that of impurity
(B) Higher melting point of the impurity than that of the pure metal
(C) Greater noble character of the solid metal than that of the impurity
(D) Greater solubility of the impurity in the molten state than in the solid

26. 50 ml of 0.2 M solution of a compound with empirical formula $CoCl_3.4NH_3$ on treatment with excess of $AgNO_3$(aq) yields 1.435 gm of AgCl. Ammonia is not removed by treatment with concentrated H_2SO_4. The formula of compound is:
(A) $\left[Co(NH_3)_4Cl_3\right]$ (B) $\left[Co(NH_3)_3Cl\right]Cl_2$
(C) $\left[Co(NH_3)_4Cl_2\right]Cl$ (D) $\left[Co(NH_3)Cl_3\right]NH_3$

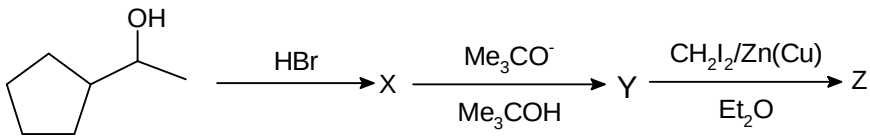
27. Determine the amount of AgI which may be dissolved in 1.0 litre of 1.0 M CN^- solution. K_{sp} for AgI and equilibrium constant for $\left[Ag(CN)_2\right]^-$ formation are $1.19\times10^{-17}M^2$ and $7.11\times10^{19}M^{-2}$ respectively.
(A) 1.0 mole (B) 0.5 mole
(C) 4.0 mole (D) 2.5 mole

Space for rough work

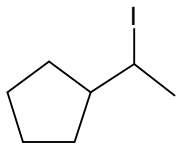
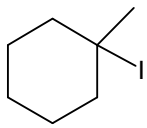
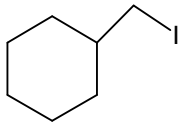
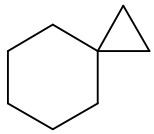
28. 

What is Z

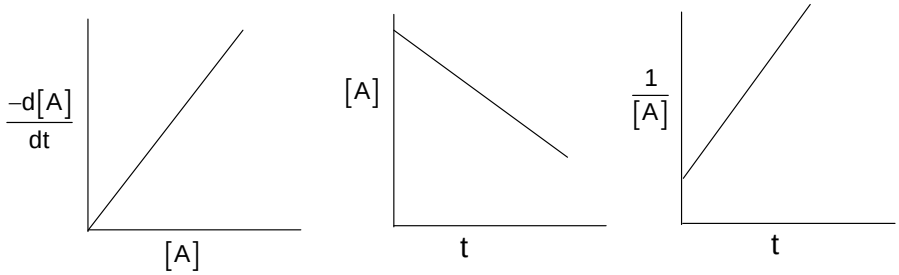
(A) D - Mannose
(B) D - Xylose
(C) D - Xylitol
(D) D - Talose

29. 

Z is

(A) 
(B) 
(C) 
(D) 

30. For a reaction $nA \longrightarrow B$, following plots are given:



The order of reaction for above plots are -

(A) 1, 0, 2
(B) 0, 1, 2
(C) 1, 1, 2
(D) 0, 2, 1

Space for rough work

31. Which of the following products are formed when potassium bromide reacts with potassium permagnate in alkaline pH?

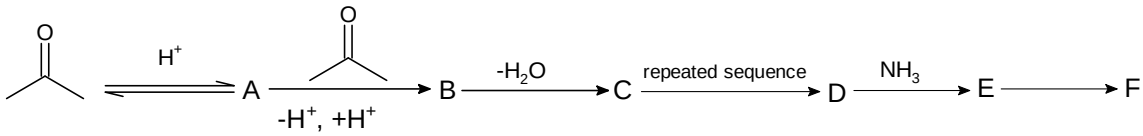
(A) BrO_3^-, MnO_2 (B) BrO_4^-, Mn^{2+}

(C) Br_2, MnO_2 (D) BrO^-, MnO_4^{2-}

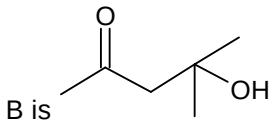
Multiple Correct Choice Type

This section contains **04 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct.

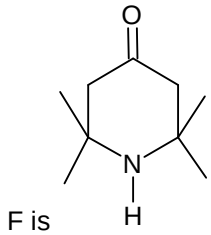
32.



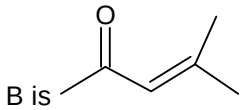
(A)



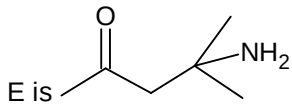
(B)



(C)



(D)



33. For an isothermal irreversible expansion of 1 mole of a perfect gas, identify the correct relation, where symbols have their usual meaning

(A) $\Delta U = 0$ (B) $\Delta H = 0$

(C) $q = RT\left(1 - \frac{P_2}{P_1}\right)$ (D) $w = RT\,\ell n\left(\frac{P_2}{P_1}\right)$

Space for rough work

34. The vander waal gas constant 'a' is given by

(A) $\frac{1}{3}V_c$ (B) $3P_cV_c^2$

(C) $\frac{1}{8}\frac{RT_c}{P_c}$ (D) $\frac{27}{64}\frac{R^2T_c^2}{P_c}$

35. Degree of hydrolysis of a salt of strong acid and weak base is:

(A) Independent of dilution (B) Increases with dilution

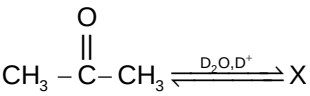
(C) Decreases with increase in K_b value (D) Increases with increase in K_b value

Comprehension Type

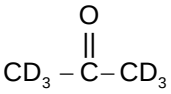
This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct.

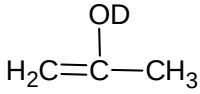
Paragraph for Questions 36 & 37

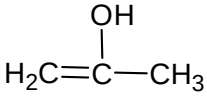
Deuterium is the isotope of hydrogen of mass number 2, with a proton and a neutron in its nucleus. The chemistry of deuterium is nearly identical to the chemistry of hydrogen, except that C – D bond is slightly (5.0 KJ/mole) stronger than the C – H bond. Reaction rates tend to be slower if a C – D bond as opposed to a C – H bond is broken in a rate limiting step. This effect on the rate is called a kinetic isotope effect.

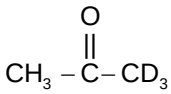
36. 

X is

(A) 

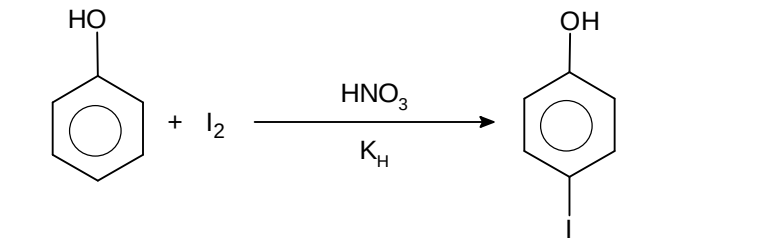
(B) 

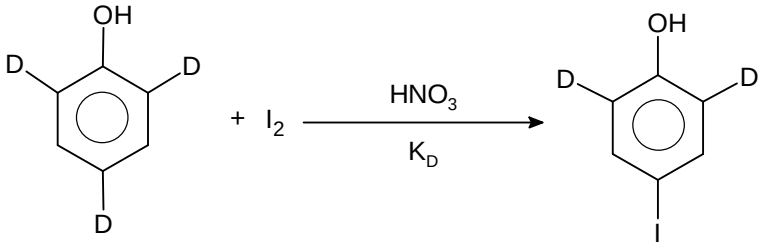
(C) 

(D) 

Space for rough work

37.





(A) $\frac{K_H}{K_D} = 1$ (B) $\frac{K_H}{K_D} \approx 4$

(C) $\frac{K_D}{K_H} \approx 4$ (D) $\frac{K_D}{K_H} \approx 2$

Paragraph for Questions 38 & 39

Galvanic cells generate electrical energy at the expense of a spontaneous redox reaction. In an electrode concentration cell two like electrodes having different concentrations, either because they are gas electrodes operating at different pressures or because they are amalgams (solutions in mercury) with different concentrations are dipped into the same solution.

Eg. An example is a cell composed of two chlorine electrodes with different pressure of Cl_2:

$Pt_L \left| Cl_2(P_L) \right| HCl(aq) \left| Cl_2(P_R) \right| Pt_R$

Where P_L and P_R are the Cl_2 pressure at the left and right electrodes.

38. Calculate the EMF of the electrode concentration cell represented by:

$Hg - Zn(c_1) \left| Zn^{+2}(aq) \right| Hg - Zn(c_2)$

At 25^0 C. c_1 = 2g of Zn per 100 g of Hg and c_2 = 1 g of Zn per 50 g of Hg

(A) 0.059 V (B) 3

(C) 0 (D) data insufficient

Space for rough work

39. A cell represented by the following cell diagram is generated to measure the solubility product of AgCl.

$Ag\left|Ag^{+}(aq)\right|Cl^{-}(aq)\left|AgCl(s)\right|Ag$

$K_{sp}^{0}(AgCl)$ at 298 K and 1 bar pressure is:

Given :

$AgCl + e^{-} \rightarrow Ag + Cl^{-}$ $\quad E^{0} = 0.22\ V$

$Ag^{+} + e^{-} \rightarrow Ag$ $\quad E^{0} = 0.79V$

$\left(\text{Take}\sqrt{10} = 3.16\ \&\ \frac{RT.(2.303)}{F} = 0.06\right)$

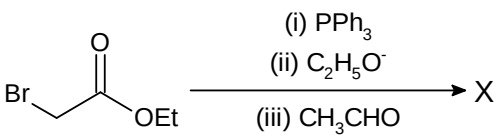
(A) 3.16×10^{-10} (B) 3.16×10^{-11}

(C) 3.16×10^{-9} (D) data insufficient

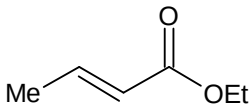
Paragraph for Questions 40 & 41

Aldehyde and Ketone molecules having α – hydrogen atoms undergo aldol condensation reaction. Aldol condensation reaction proceeds in acidic and basic conditions. In basic medium, aldol condensation product will be formed involving enolate ion formation. Here the enolate ion acts as nucleophile. Other carbonyl group containing compounds also perform similar type of reactions.

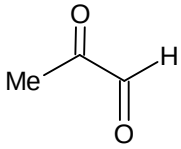
40.



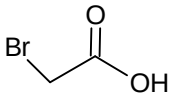
Final product X is

(A) 

(B) CH_3CHO and CH_3COOH

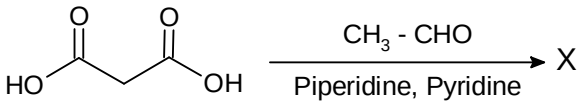
(C) 

and C_2H_5OH

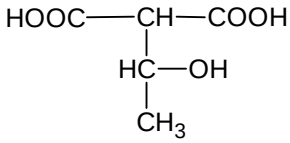
(D) 

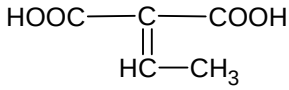
and C_2H_5OH

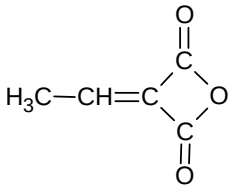
Space for rough work

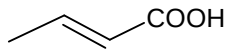
41. 

Final product X is

(A) 

(B) 

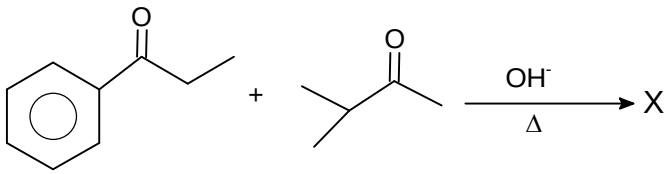
(C) 

(D) 

SECTION – C
(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

42. Let (+) tartaric acid has a specific rotation of + 11.1. Calculate the specific rotation of a mixture of 68% (+) tartaric acid and 32% (-) tartaric acid.

43. 

Total number of π bonds in X is

44. Find the number of molecules having two lone pair of electrons on central atom.
$I_3^+, XeF_2, XeF_4, H_2O, NH_2^-, H_2S, H_2SO_4, NF_3$

45. A Boron mineral have pentaborate anion whose molecular formula is $\left[B_5O_6(OH)_4\right]^-$. The number of B – O - B linkage present are?

46. The electron in the first excited state ($n_1 = 2$) of H – atom absorbs a photon and is further excited to n^{th} shell. The de-Broglie wavelength of the electron in this excited state is 1340 pm. Find the value of n.

Space for rough work

Mathematics

PART – III

SECTION – A
Straight Objective Type

This section contains **08 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct.

47. The sum of abscissa of the points of inflexions of curve y = $f(x) = \frac{x^7}{42} - \frac{2x^6}{15} + \frac{3x^5}{20} + \frac{x^4}{3} - \frac{2x^3}{3} + 99x + 2010$ is,

(A) -1 (B) 0
(C) 2 (D) none of these

48. In triangle ABC, if y = x+1 & y = 2x + 3 are altitude through A & angle bisector of B respectively. If vertex C ≡ (3 , 4) then equation of median through C is

(A) x + 7y = 25 (B) 7x + 2y = 25
(C) 7x + y = 25 (D) 2x + 7y = 25

49. Equation of curve passing through (1, 1) & satisfying the differential equation $(xy^2 + y^2 + x^2y + 2xy)dx + (2xy + x^2)dy = 0$ Is

(A) $xy(x+y) = 2e^{1-x}$ (B) $xy(x^2 + y^2) = 2e^{x-1}$
(C) $x^2y^2(x+y) = 2$ (D) $x^2y^2(x^2 + y^2) = 2$

50. In a bag there are 10 black & 10 white balls. A ball is drawn at random & 5 extra balls of same color as of drawn ball are added in the bag along with the drawn ball. Now another ball is drawn and replaced in the bag but 4 balls of color same as drawn ball are removed from the bag. Again a ball is drawn and found to be white find the probability that the second drawn ball was black.

(A) $\frac{4}{7}$ (B) $\frac{3}{7}$
(C) $\frac{2}{7}$ (D) $\frac{1}{7}$

Space for rough work

51. Let $f(x) = x + \{x\}^2$ where {x} is fractional part of x. The area between the curves y = f(x), $y = f^{-1}(x)$, x = 0 and x = $\frac{3}{2}$ is,

(A) $\frac{1}{2}$ (B) 1

(C) $\frac{3}{2}$ (D) 2

52. f & g are two real valued continuous functions and let $\int g(x)dx = f^{-1}(x) and\, f(x) = x^3 + x + \sin \pi x + 2$ then the value of $\int_2^4 xg(x)dx$ is

(A) $\frac{-11}{4} - \frac{2}{\pi}$ (B) $\frac{11}{4} - \frac{2}{\pi}$

(C) $\frac{-11}{4} + \frac{2}{\pi}$ (D) $\frac{11}{4} + \frac{2}{\pi}$

53. The value of the limit $\lim\limits_{x \to \frac{1}{\sqrt{2}}^+} \dfrac{\cos^{-1}\left(2x\sqrt{1-x^2}\right)}{x - \frac{1}{\sqrt{2}}}$ is equal to

(A) $\sqrt{2}$ (B) $-2\sqrt{2}$

(C) $2\sqrt{2}$ (D) doesn't exist

54. If $\left[1 + x + x^2\right]^n = \sum_{r=0}^{2n} a_r x^r$, then the value of $\sum_{k=0}^{3r} (-1)^k\, a_k\, {}^nC_{3r-k}$ $(where\ 3r \le n)$ *is* $a_1^{a_2}C_{a_3}$. Then value of $a_1 + a_2 + a_3$ is

(A) n+1 (B) r+1

(C) n+r+1 (D) n + r

Space for rough work

Multiple Correct Choice Type

This section contains **04 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct.

55. $x + y = 2$ and $x - y = 2$ are tangents on a parabola at (1, 1) and (4, 2) respectively. Which of the followings is/are correct.

(A) Equation of directrix is $x + 3y = 2$ (B) Equation of axis is $3x - y = 5$

(C) Focus of the parabola is at $\left(\frac{8}{5},\frac{6}{5}\right)$ (D) Vertex of the parabola is at $\left(\frac{33}{20},\frac{13}{20}\right)$

56. Line OQ is angle bisector of angle O of right angle triangle OPR, right angled at P. Point Q is such that ORQP is concyclic. If point O is origin and points P, Q, R are represented by the complex numbers z_3, z_2, z_1 respectively. If $\frac{z_2^{\,2}}{z_1 z_2}=\frac{3}{2}$ then (R is circum radius of ΔOPR)

(A) Angles of ΔOPR are $\frac{\pi}{6},\frac{\pi}{3},\frac{\pi}{2}$ (B) Angles of ΔOPR are $\frac{\pi}{4},\frac{\pi}{4},\frac{\pi}{2}$

(C) Area of ΔOPR is $2\sqrt{2}R^2$ (D) Area of ΔOPR is $2R^2$

57. In ΔABC, X and Y are foot of perpendicular from A and C respectively on median BE. If $\frac{area(\Delta BYC)}{area(\Delta AXE)}=4$, then which of the following is/are correct.

(A) $\frac{area(\Delta ABC)}{area(\Delta AXB)}=3$ (B) BX = 2XY

(C) 2 area (ΔAXB) = area (ΔXYC) (D) none of these

Space for rough work

58. If a, b, c are in AP and A, B, C are in GP (common ratio $\neq 1$). Then which of the following is/are correct.

(A) $\frac{A}{a},\frac{B}{b},\frac{C}{c}$ are in HP if common ratio of GP is c/a.

(B) $\frac{a}{A},\frac{b}{B},\frac{c}{C}$ are in HP if common ratio of GP is equal to common difference of AP.

(C) $\frac{A^2}{a},\frac{B^2}{b},\frac{C^2}{c}$ are in HP if common ratio of GP is $\sqrt{\frac{c}{a}}$

(D) $\frac{a}{A^2},\frac{b}{B^2},\frac{c}{C^2}$ are in HP if common ratio of GP is equal to square root of common difference of AP.

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct.

Paragraph for Questions 59 & 60

Lines $L_1 : \frac{x-6}{3} = \frac{y-4}{2} = z-2$ and $L_2 : \frac{x-8}{4} = y-2 = \frac{z-4}{2}$ meets the plane $\pi : \vec{r}.\left(2\hat{I}+\hat{j}-\hat{k}\right) = 7$ at points A and B.

59. Area of the triangle formed by the lines L_1, L_2 & AB is

(A) $\frac{7}{2}\sqrt{\frac{19}{2}}$ (B) $\sqrt{\frac{19}{2}}$

(C) $\frac{\sqrt{11}}{4}$ (D) $6\sqrt{5}$

Space for rough work

60. Equation of line joining the images of the point of intersection of L_1 & L_2 with respect to the plane π and the line AB is

(A) $\frac{3x-14}{4}=\frac{3y-7}{5}=\frac{3z+7}{1}$ (B) $\frac{3x-14}{3}=\frac{3y-7}{3}=\frac{3z+7}{6}$

(C) $\frac{3x-14}{0}=\frac{3y-7}{3}=\frac{3z+7}{4}$ (D) $\frac{3x-14}{0}=\frac{3y-7}{3}=\frac{3z+7}{3}$

Paragraph for Questions 61 & 62

Let $x_i \in R$, $i = 1,2,3.....n$ are numbers such that $\sum_{i=1}^{n} i\sqrt{x_i - i^2} = \frac{\sum_{i=1}^{n} x_i}{2}$ and $x_1 + x_2 + ... + x_n = 280$

61. No. of ways of distribution of n identical objects among 3 persons such that each get at least 1 object is

(A) 4 (B) 10

(C) 20 (D) 140

62. Probability that a randomly selected triangle formed by vertices of a 2n + 1 sided regular polygon is isosceles is

(A) $\frac{3}{13}$ (B) $\frac{5}{13}$

(C) $\frac{7}{13}$ (D) $\frac{9}{13}$

Paragraph for Questions 63 & 64

Let p (n) denotes the number of different ways the positive integer n (n>1) can be expressed as the sum of 1's and 2's. For example P (5) = 8, i – e

i.e. 5 =1 + 1 + 1 + 1 +1

=1 + 1 + 1 + 2 = 1+ 1+ 2+1 = 1 + 2 + 1 + 1 = 2 + 1 + 1 + 1

=1 + 2 + 2 = 2 + 1 + 2 =2 + 2 + 1

63. The value of p(p(6)) is

(A) 356 (B) 377

(C) 389 (D) 427

Space for rough work

64. The number of solutions of the equation p(n) = n + 1 where n = N is
(A) 1 (B) 2
(C) 3 (D) 4

SECTION – C
(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

65. Number of triangle ABC $\angle B = 90^0$ such that point B is vertex and A & C are point of zeros of a Quadratic expression $y = ax^2 + bx + c$ where b is an odd integer & $a, c \in z$, is

66. The value of $\left[\int_{-\pi}^{\pi}\sqrt{\frac{|\sin x|}{1+\tan^2 x}}dx\right]$ is (where [.] is greatest integer).

67. Two circles of unequal radii have four common tangents. A transverse common tangent meets the direct common tangents at the points P & Q. If length of direct tangent (between the point of contacts) is 8 then length of PQ is

68. If $A=\begin{bmatrix}5 & -6\\ 1 & -1\end{bmatrix}$ then the value of $\frac{\det\left(A^m-5A^{m-1}\right)}{\det\left(A^n-5A^{n-1}\right)}\left(m,n\in N\right)is$

69. Integral part of the area of figure bounded by the tangents at the end of latus rectum of ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ and directrices of hyperbola $\frac{x^2}{9}-\frac{y^2}{72}=1$ is

Space for rough work